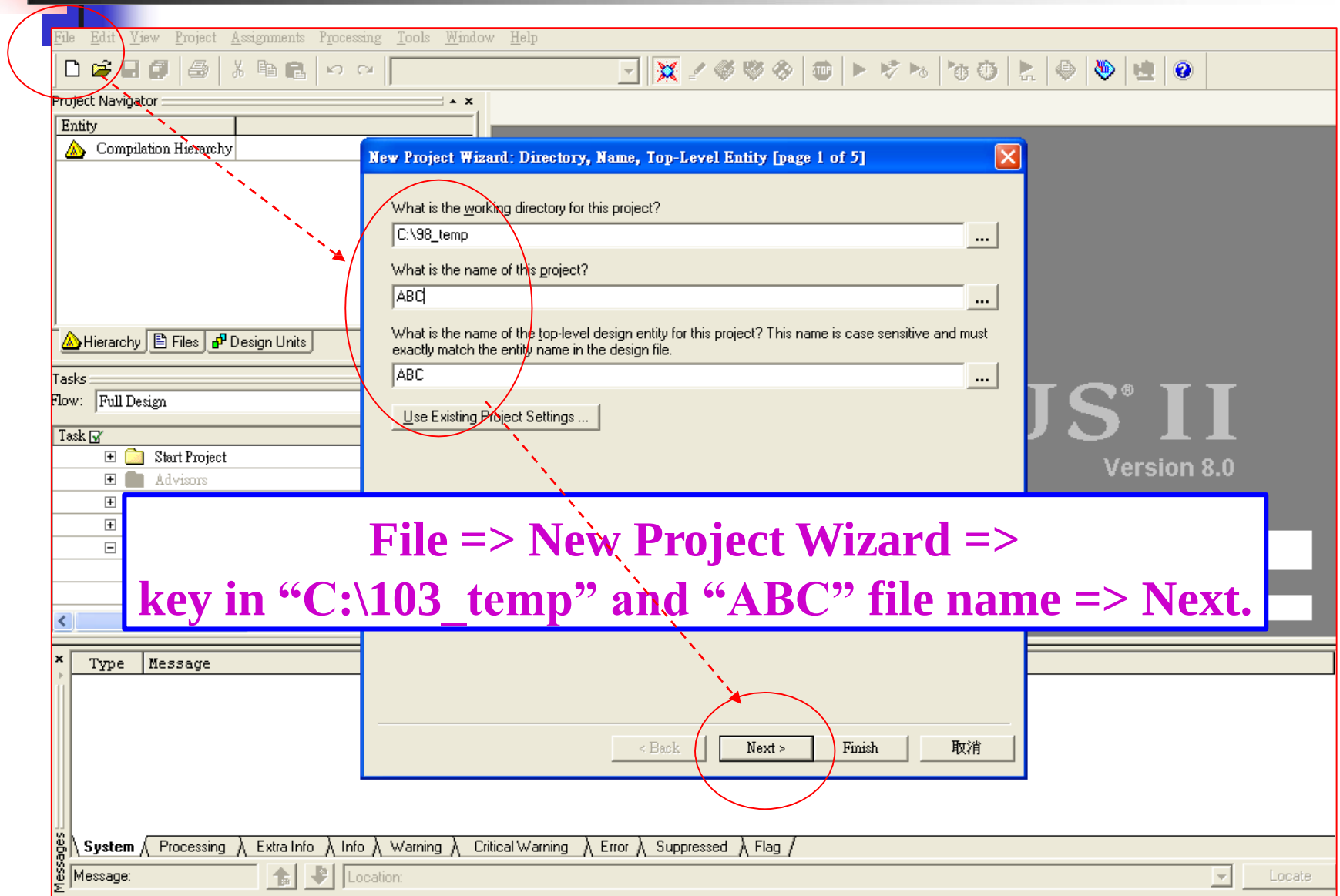


Verilog 硬體描述語言

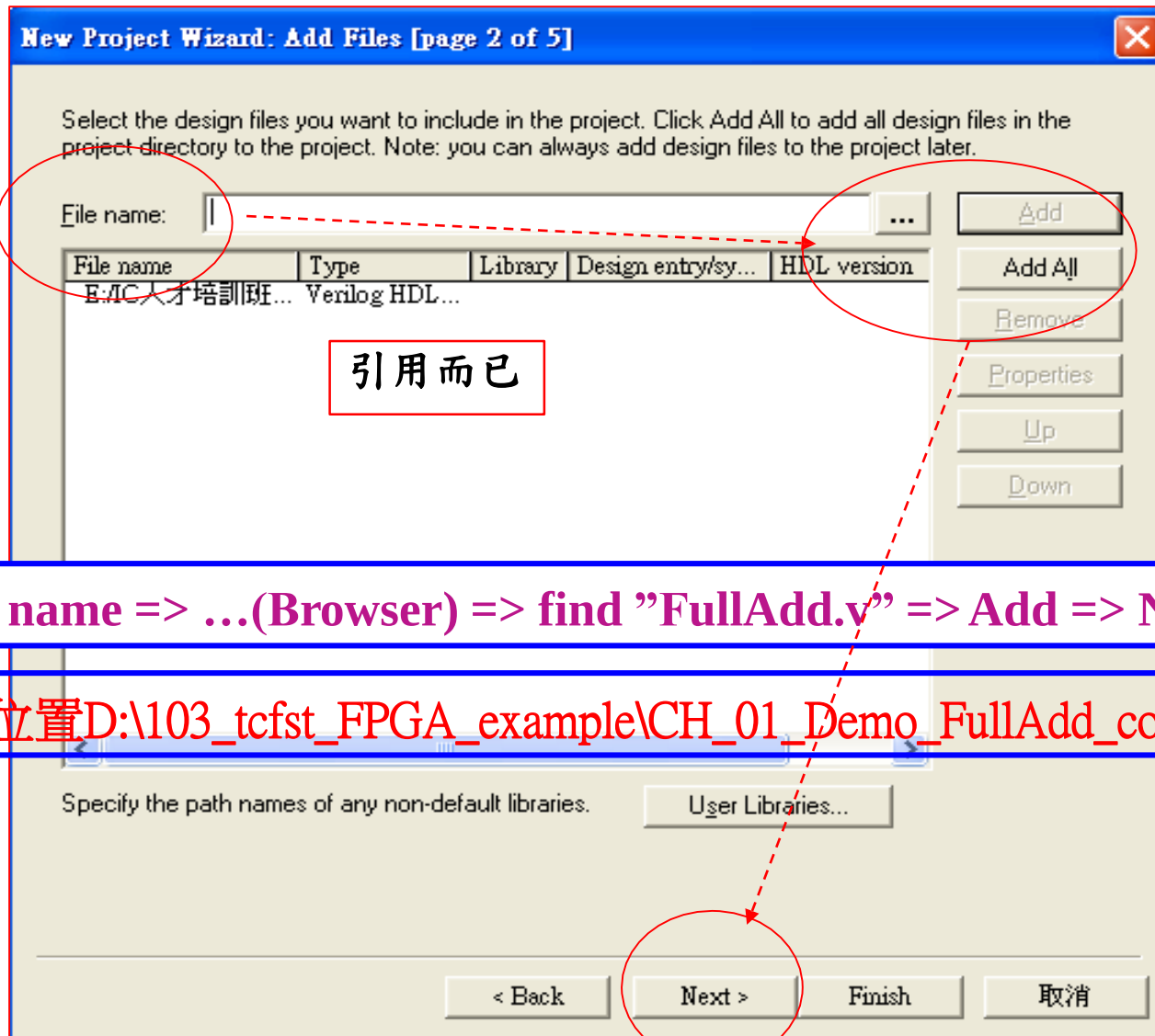
Quatus II 9.1

使用練習

Quartus II 使用練習 (2)



Quartus II 使用練習 (3)



Quartus II 使用練習 (4)

New Project Wizard: Family & Device Settings [page 3 of 5]

Select the family and device you want to target for compilation.

Device family
Family: Cyclone
Devices: All

Target device
☐ Auto device selected by the Fitter
☒ Specific device selected in 'Available devices' list

Show in 'Available device' list
Package: Any
Pin count: 240
Speed grade: 8
☒ Show advanced devices
☐ HardCopy compatible only

Available devices:

Name	Core v...	LEs	Memor...	PLL
EP1C6Q240C8	1.5V	5980	92160	2
EP1C12Q240C8	1.5V	12060	239616	2

Cyclone => "240" => "8" => (first item) => Next.

Companion device
HardCopy:
☒ Limit DSP & RAM to HardCopy device resources

< Back Next > Finish 取消

Quartus II 使用練習 (5)

New Project Wizard: EDA Tool Settings [page 4 of 5]

Specify the other EDA tools -- in addition to the Quartus II software -- used with the project.

Design Entry/Synthesis

Tool name:

Format:

☐ Run this tool automatically to synthesize the current design

Simulation

Tool name:

Format:

☐ Run gate-level simulation automatically after compilation

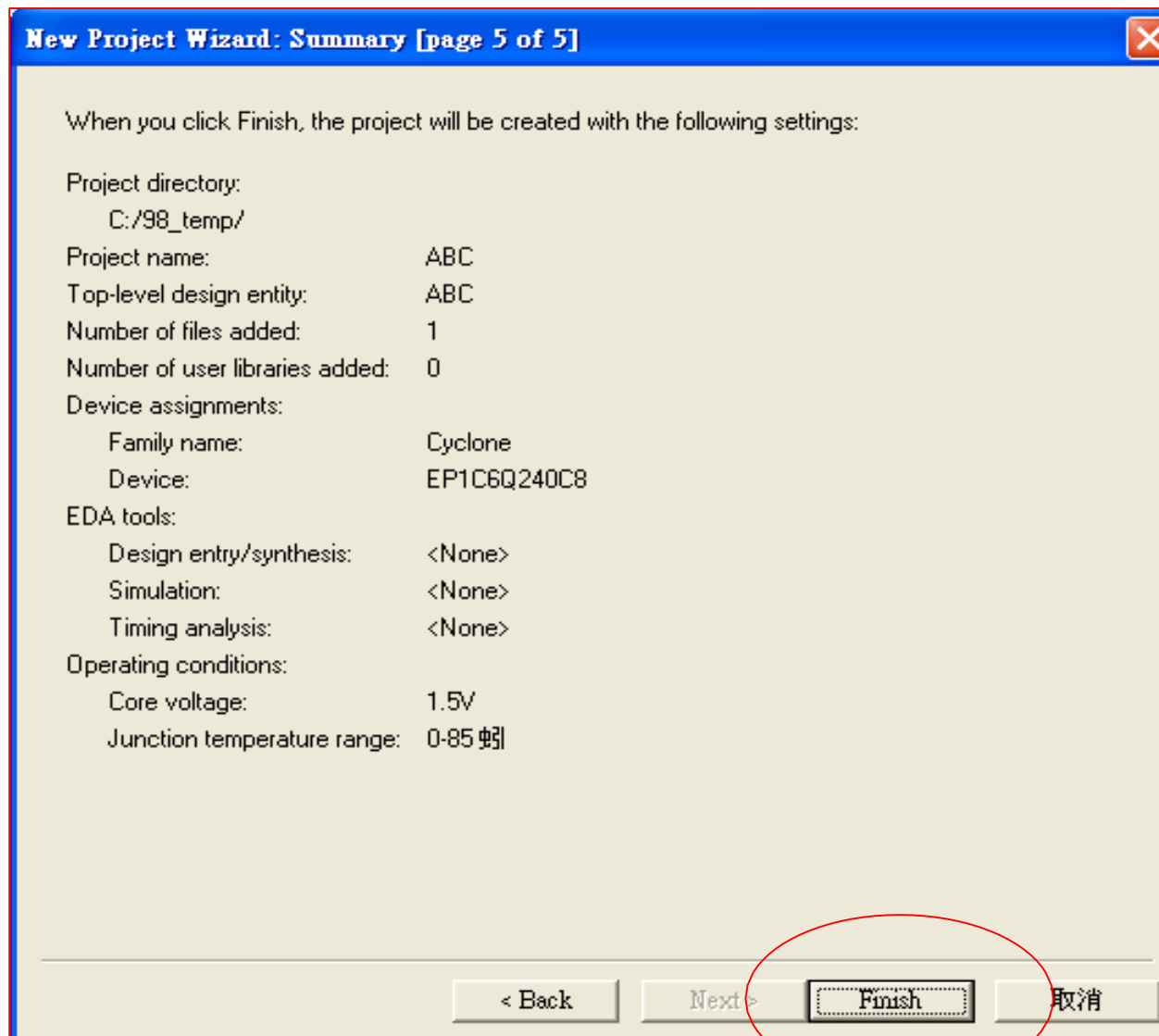
Timing Analysis

Tool name:

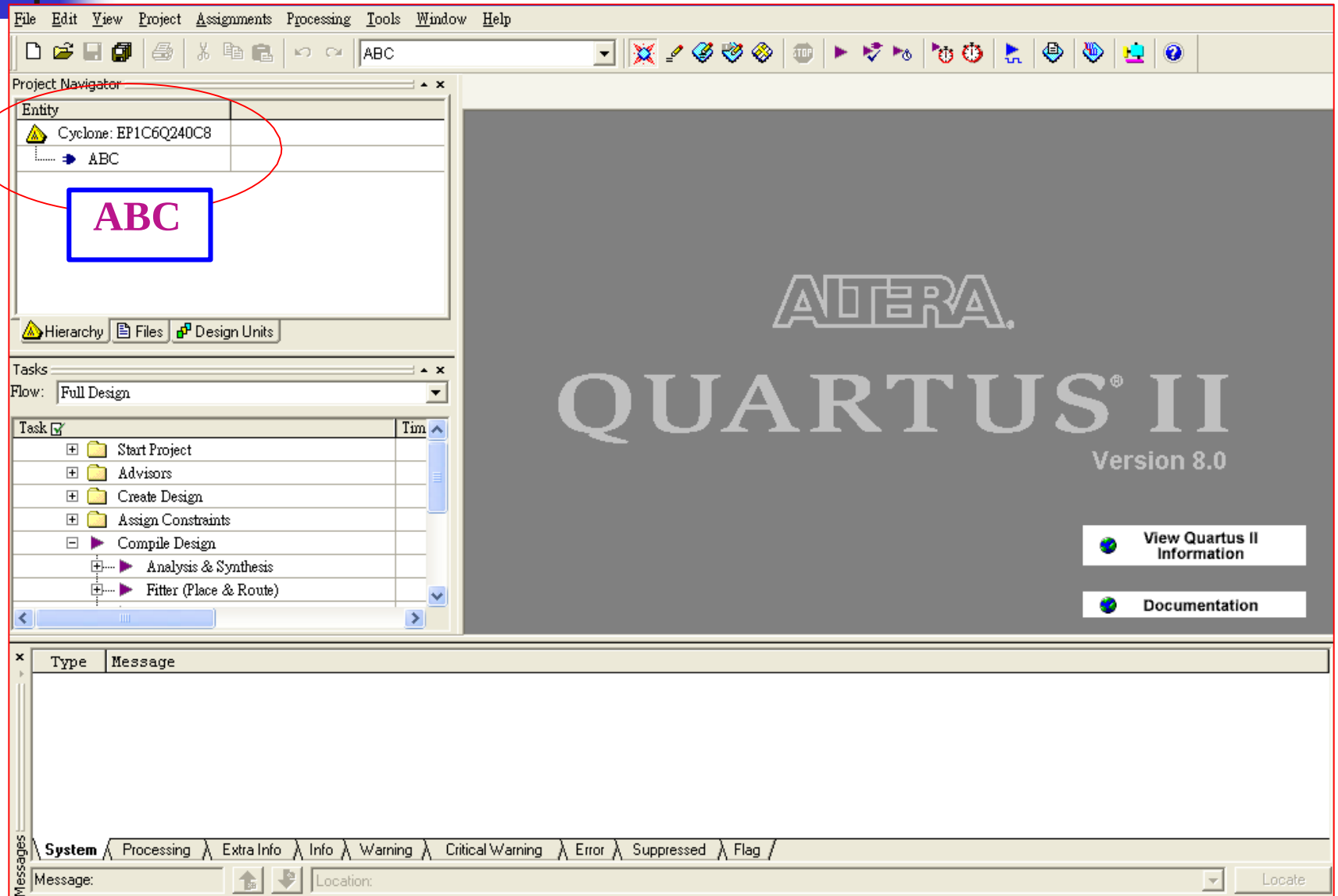
Format:

☐ Run this tool automatically after compilation

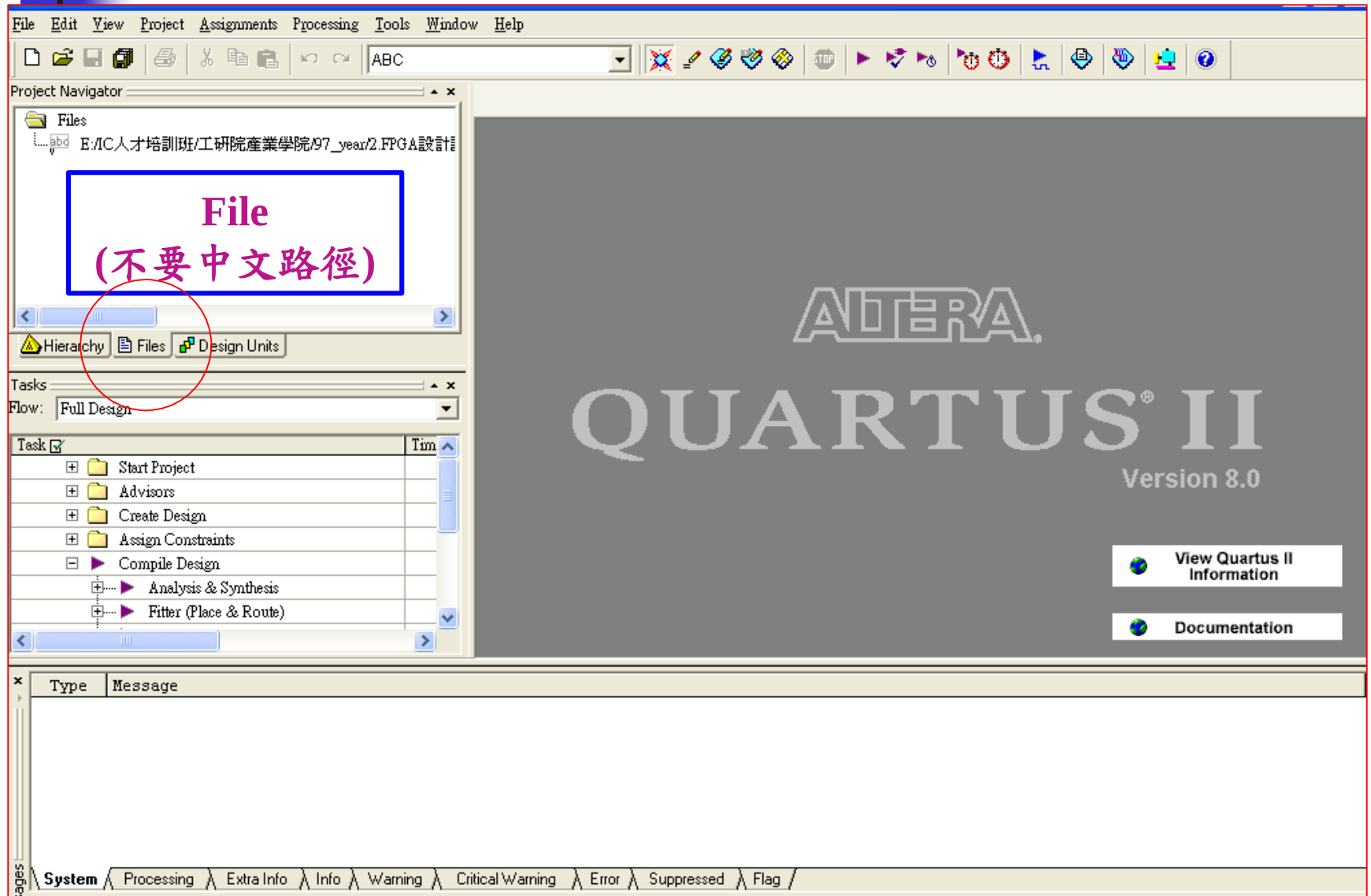
Quartus II 使用練習 (6)



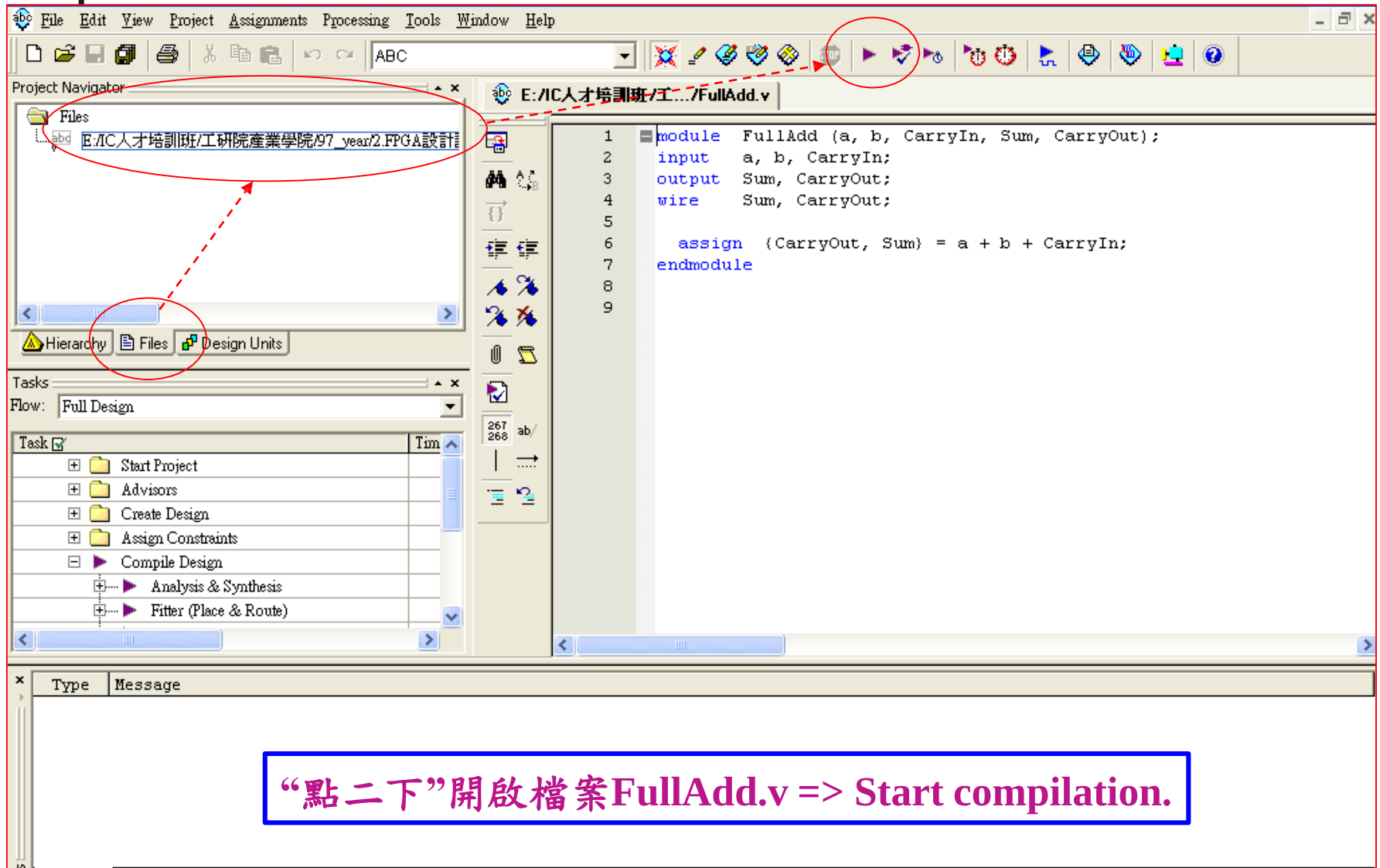
Quartus II 使用練習 (7)



Quartus II 使用練習 (8)



Quartus II 使用練習 (9)



Quartus II 使用練習 (10)

The screenshot shows the Quartus II IDE with the 'Flow Summary' window open. The 'Flow Status' is 'Flow Failed - Sat Apr 18 18:16:10 2009'. The 'Flow Summary' table lists various components and their status:

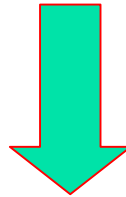
Flow Status	Flow Failed - Sat Apr 18 18:16:10 2009
Quartus II Version	8.0 Build 215 05/29/2008 SJ Full Version
Revision Name	ABC
Top-level Entity Name	ABC
Family	Cyclone
Device	EP1C6Q240C8
Timing Models	Final
Timing requirements	N/A
Logic elements	N/A until Partition Merge
IO pins	N/A until Partition Merge
Memory bits	N/A until Partition Merge
9-bit elements	N/A until Partition Merge
...	N/A until Partition Merge
...	N/A until Partition Merge

A dialog box titled 'Quartus II' is displayed, stating: 'Full Compilation was NOT successful (3 errors)'. The dialog has a '確定' (OK) button. A green arrow points from the dialog to a text box at the bottom.

**Compilation NOT successful => “確定” => 原因是
模組名單與檔名不同，出現編譯上的錯誤。
=>修正“檔案名稱”或修正“模組名稱”。
=> Set as Top-level Entry**

Quartus II 使用練習 (11)

Type	Message
+	Info: Running Quartus II Analysis & Synthesis
+	Info: Command: quartus_map --read_settings_files=on --write_settings_files=off ABC -c ABC
+	Info: Found 1 design units, including 1 entities, in source file E:/IC人才培訓班/工研院產業學院/97_year/2.FPGA設計語法介紹: verilog
✗	Error: Top-level design entity "ABC" is undefined
+	Error: Quartus II Analysis & Synthesis was unsuccessful. 1 error, 0 warnings
✗	Error: Quartus II Full Compilation was unsuccessful. 3 errors, 0 warnings



❑ Error: Top-level design entity "ABC" is undefined.

⇒ 出現在階層內” Top-level “的單體” ABC”未被定義。

⇒ 必須修改單體（entity）的名稱或模組(module)的名稱。

❑ Error: Quartus II Analysis & Synthesis was unsuccessful. 1 error, 0 warnings

❑ Error: Quartus II Full Compilation was unsuccessful. 1 error, 0 warnings

Quartus II 使用練習 (12)

The screenshot shows the Quartus II IDE with the following components:

- Project Navigator:** Displays the project structure. The file `E://ic人才培訓班/工研院產業學院/97_year/2.FPGA設計` is selected.
- Tasks:** A list of tasks for the project. The tasks are: Start Project, Advisors, Create Design, Assign Constraints, Compile Design (00:0), Analysis & Synthesis (00:0), and Fitter (Place & Route). The Compile Design task is highlighted with a red 'X'.
- Code Editor:** Displays the Verilog code for the module `ABC`. The code is as follows:

```
1 module ABC(a, b, CarryIn, Sum, CarryOut);  
2   input  a, b, CarryIn;  
3   output Sum, CarryOut;  
4   wire   Sum, CarryOut;  
5  
6   assign {CarryOut, Sum} = a + b + CarryIn;  
7 endmodule  
8  
9
```

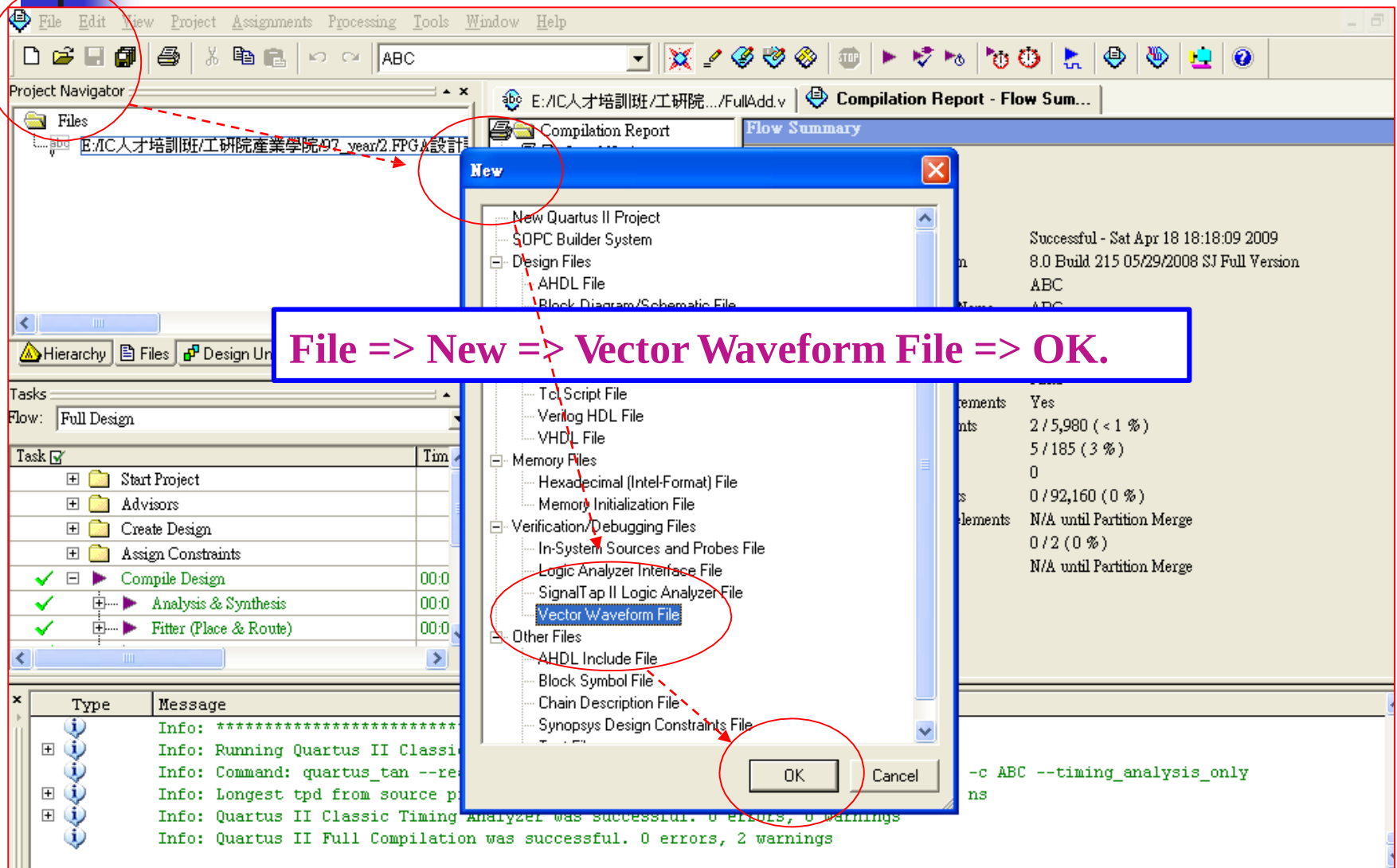
Select “E://ic人才.../FullAdd.v* => 修正”模組名稱”為”ABC” => Save => Compilation。

Quartus II 使用練習 (13)

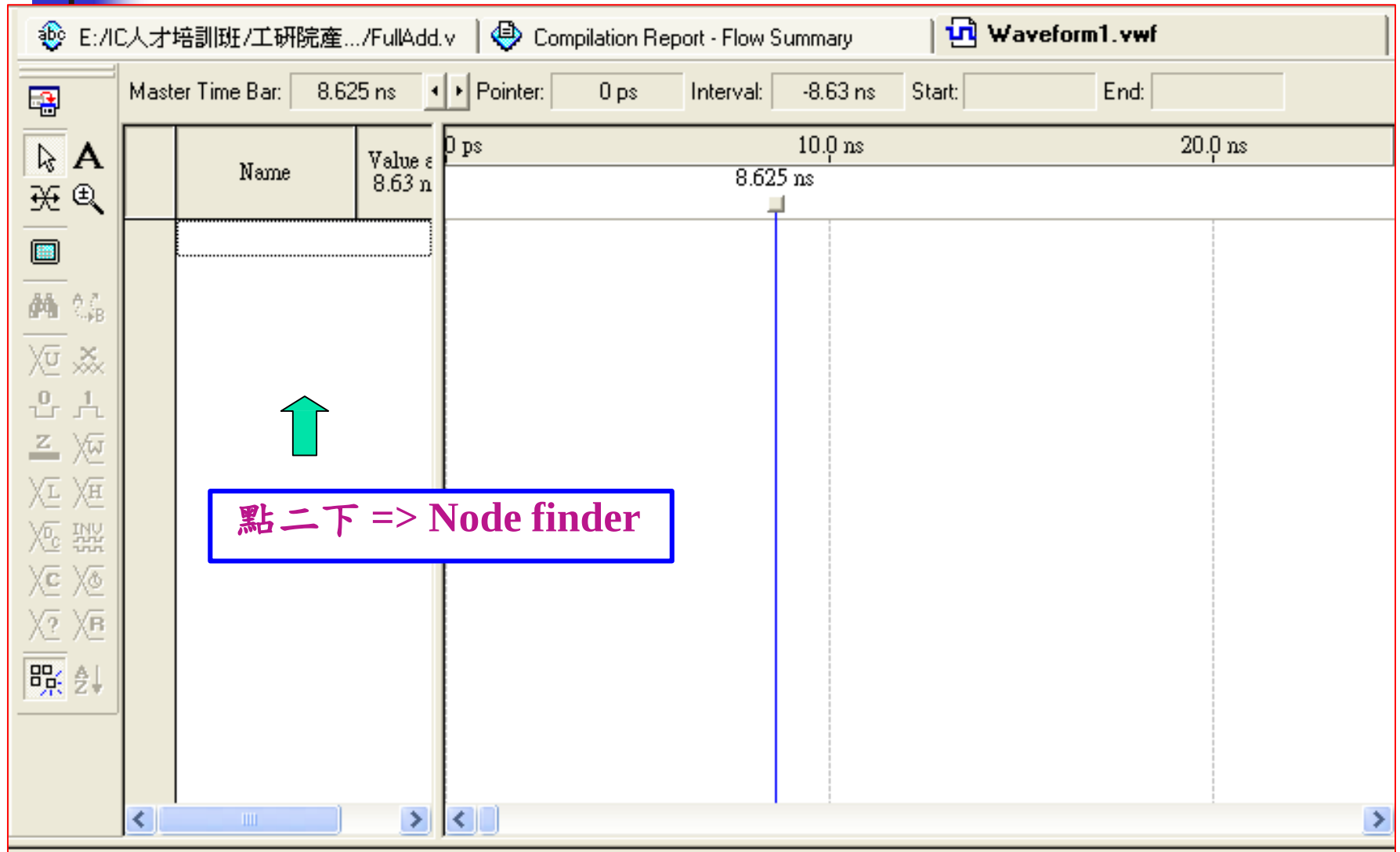
The screenshot displays the Quartus II software interface during a successful compilation process. The main window shows the 'Flow Summary' for a project named 'ABC'. The flow status is 'Successful - Sat Apr 18 18:18:09 2009'. The summary includes details such as the Quartus II version (8.0 Build 215 05/29/2008 SJ Full Version), revision name (ABC), top-level entity name (ABC), family (Cyclone), device (EP1C6Q240C8), and timing models (Final). A dialog box titled 'Quartus II' is overlaid on the center, indicating 'Full Compilation was successful (2 warnings)' with a '確定' (OK) button. The 'Tasks' pane on the left shows the 'Full Design' flow with the following tasks completed: 'Start Project', 'Advisors', 'Create Design', 'Assign Constraints', 'Compile Design' (00:0), 'Analysis & Synthesis' (00:0), and 'Fitter (Place & Route)' (00:0). The 'Messages' pane at the bottom shows the following log entries:

```
Info: *****
Info: Running Quartus II Classic Timing Analyzer
Info: Command: quartus_tan --read_settings_files=off --write_settings_files=off ABC -c ABC --timing_analysis_only
Info: Longest tpd from source pin "CarryIn" to destination pin "CarryOut" is 11.003 ns
Info: Quartus II Classic Timing Analyzer was successful. 0 errors, 0 warnings
Info: Quartus II Full Compilation was successful. 0 errors, 2 warnings
```

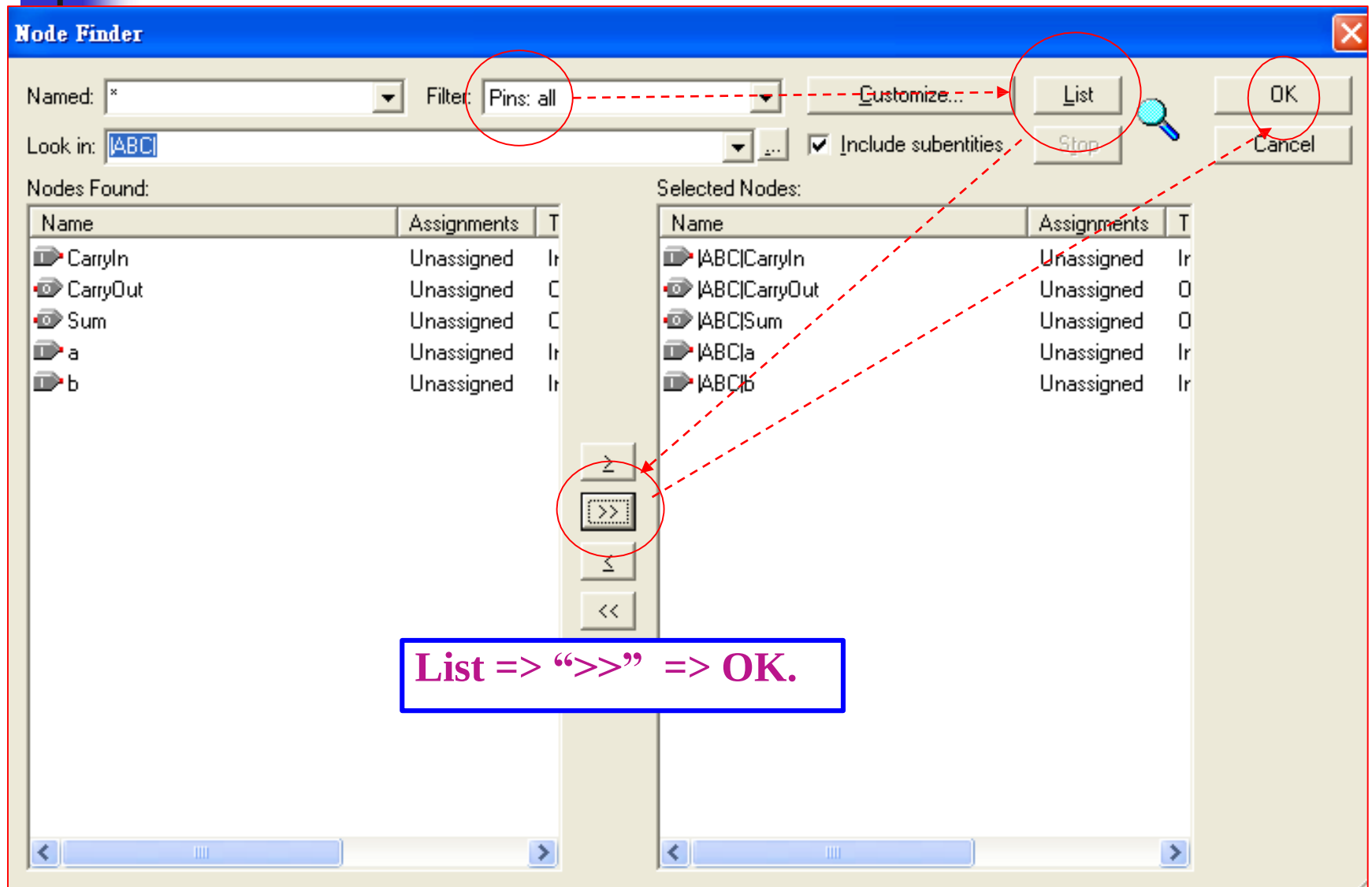
Quartus II 使用練習 (14)



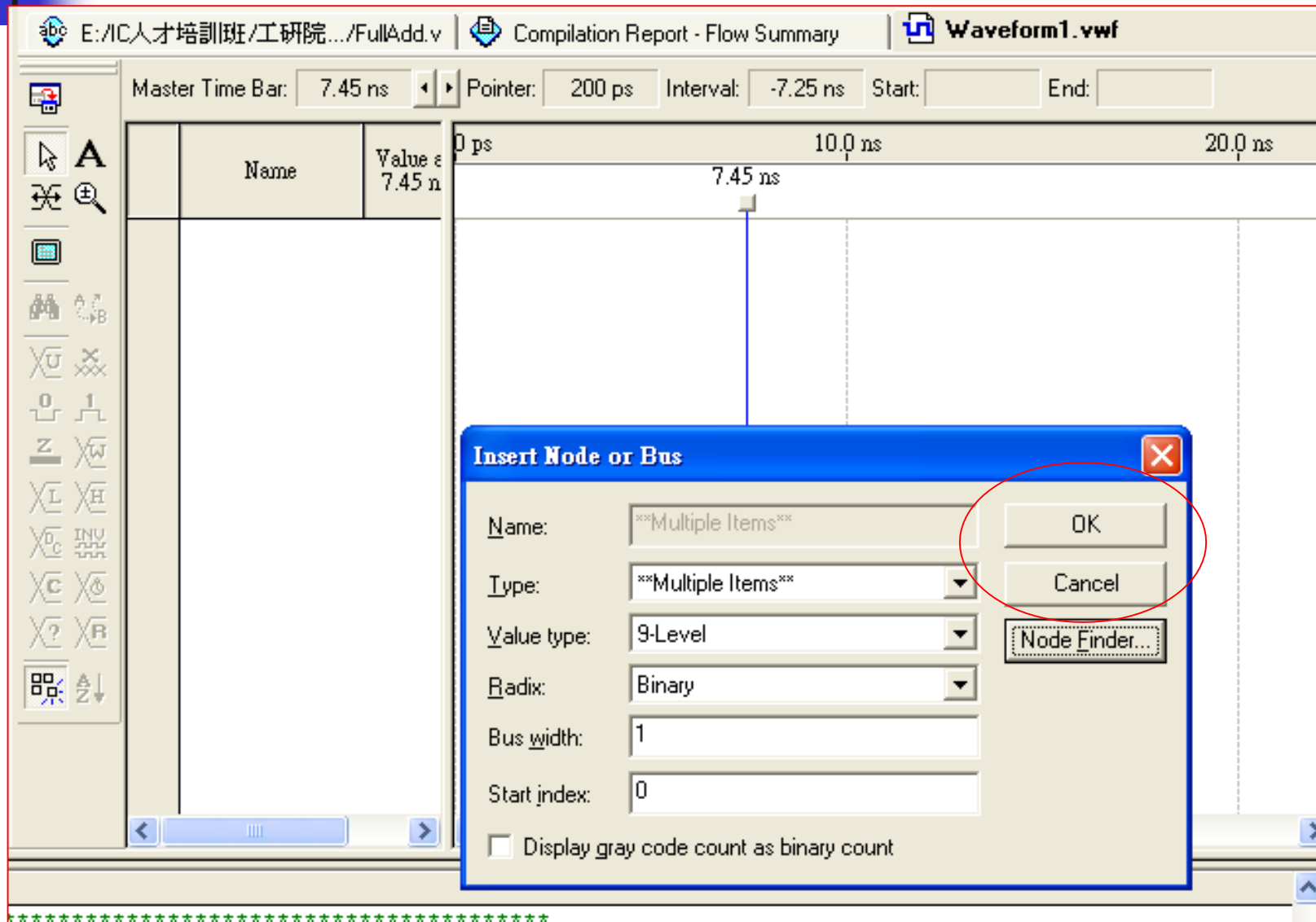
Quartus II 使用練習 (15)



Quartus II 使用練習 (16)



Quartus II 使用練習 (17)



Quartus II 使用練習 (18)

The screenshot shows the Quartus II interface with the Timing Analyzer window open. The Timing Analyzer window displays a list of signals: CarryIn, CarryOut, Sum, a, and b. The signal 'a' is highlighted in blue. The Master Time Bar shows 7.45 ns. The Pointer is at 350 ps. The Interval is -7.1 ns. The Start is 0 ps and the End is 1.0 us. The waveform shows a clock signal for 'a' with a period of 10.0 ns and a duty cycle of 50%.

The Clock dialog box is open, showing the following settings:

- Time range: Start time: 0 ps, End time: 1.0 us
- Base waveform on: ☐ Clock settings: ☐ Time period: ☒
- Period: 10.0 ns
- Offset: 0.0 ns
- Duty cycle (%): 50

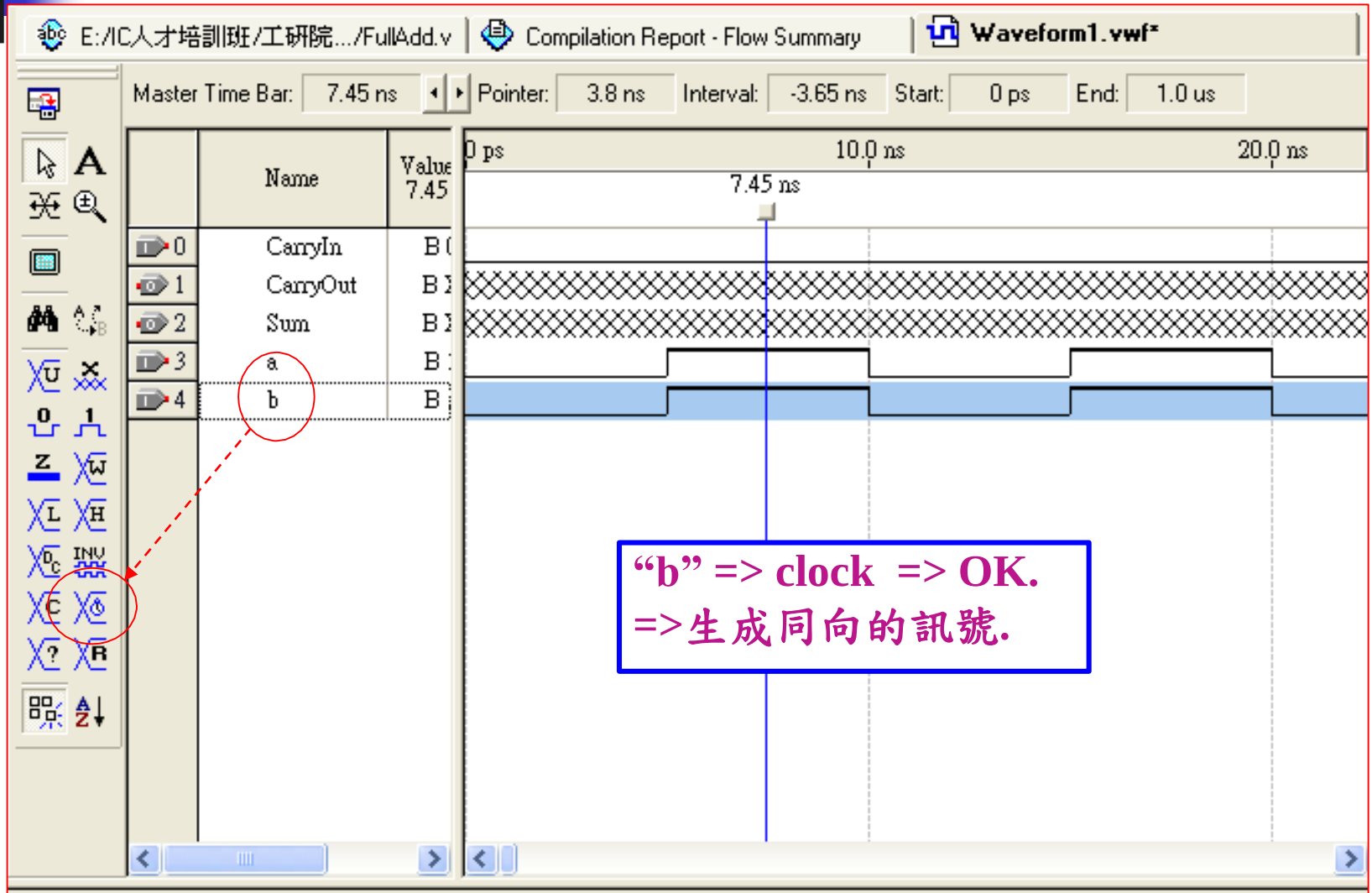
The OK button is highlighted with a red circle. A red dashed line points from the 'a' signal in the Timing Analyzer window to the 'Time period' radio button in the Clock dialog box.

“a” => clock => OK.

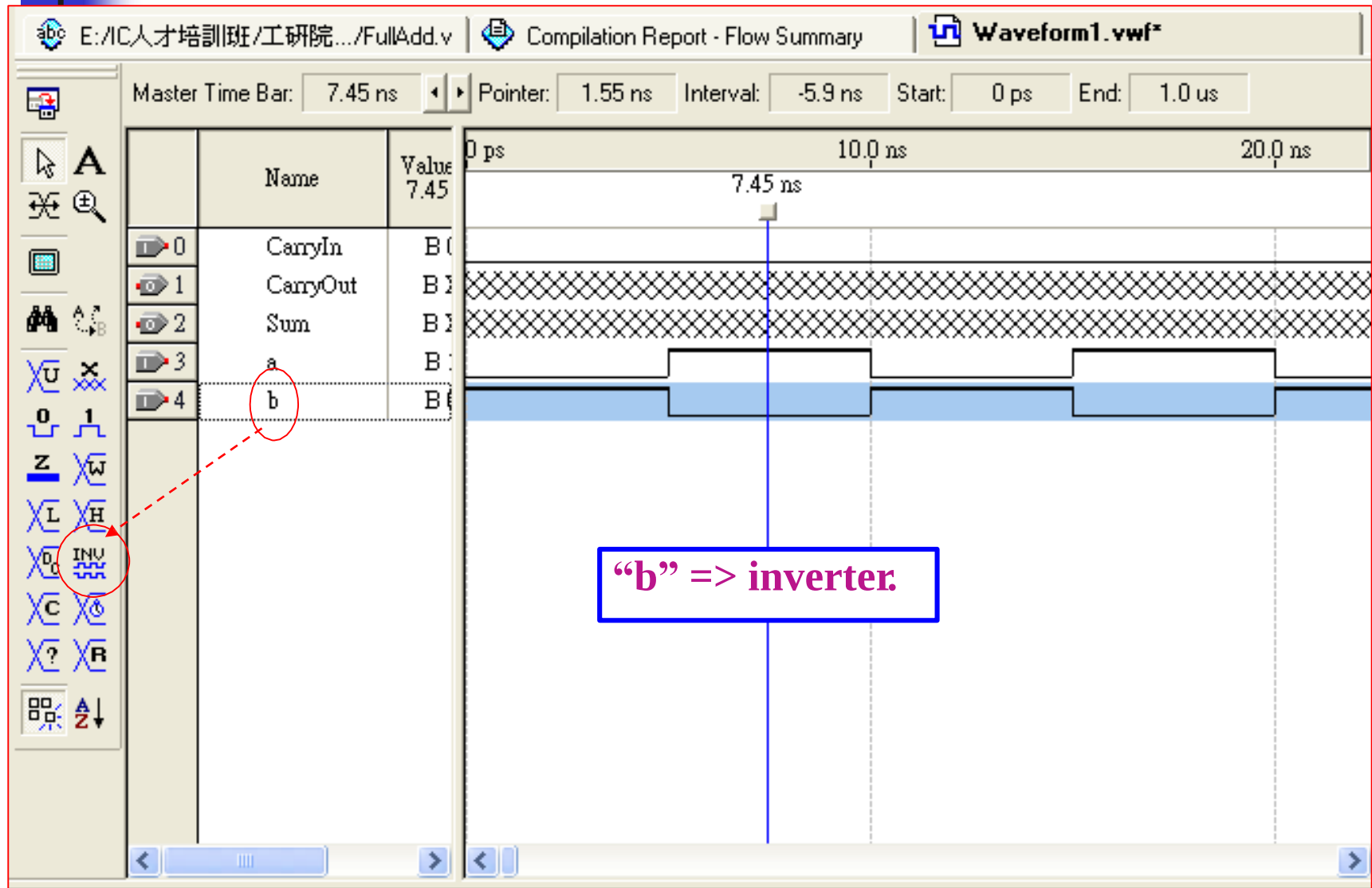
Timing Analyzer
settings_files=off
"CarryIn" to desti
alyzer was successf
was successful. 0 e

BC --timing_analysis_only

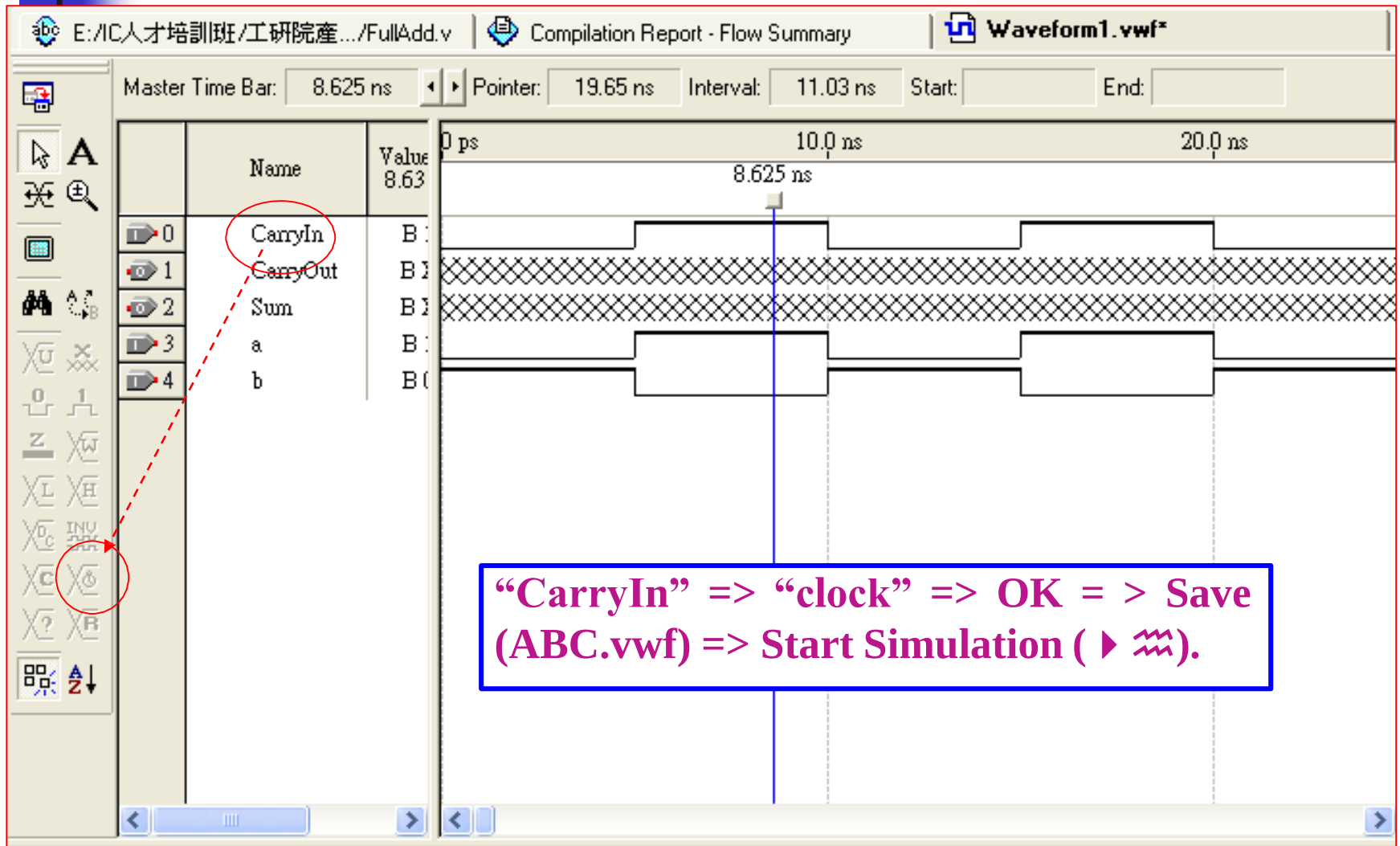
Quartus II 使用練習 (19)



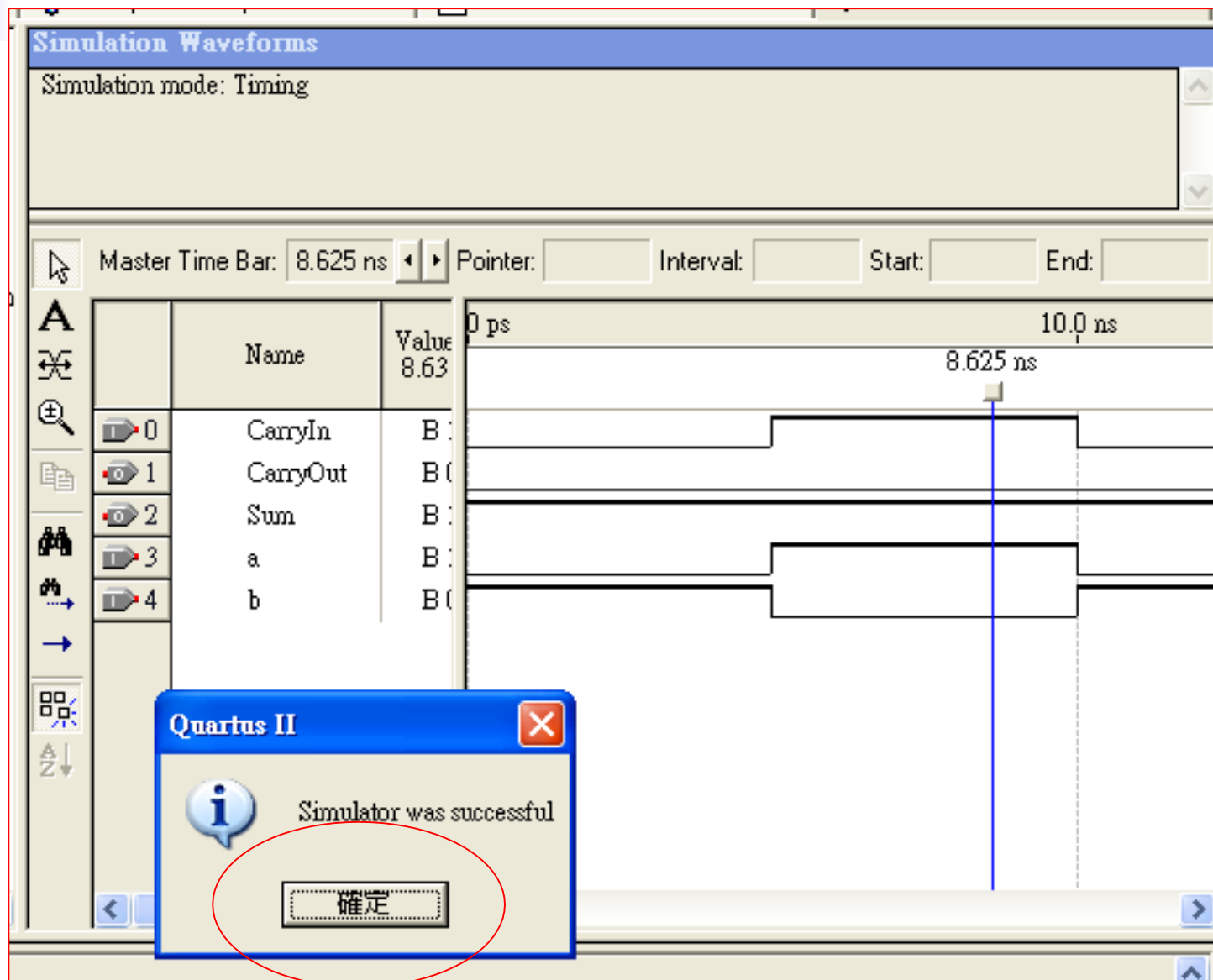
Quartus II 使用練習 (20)



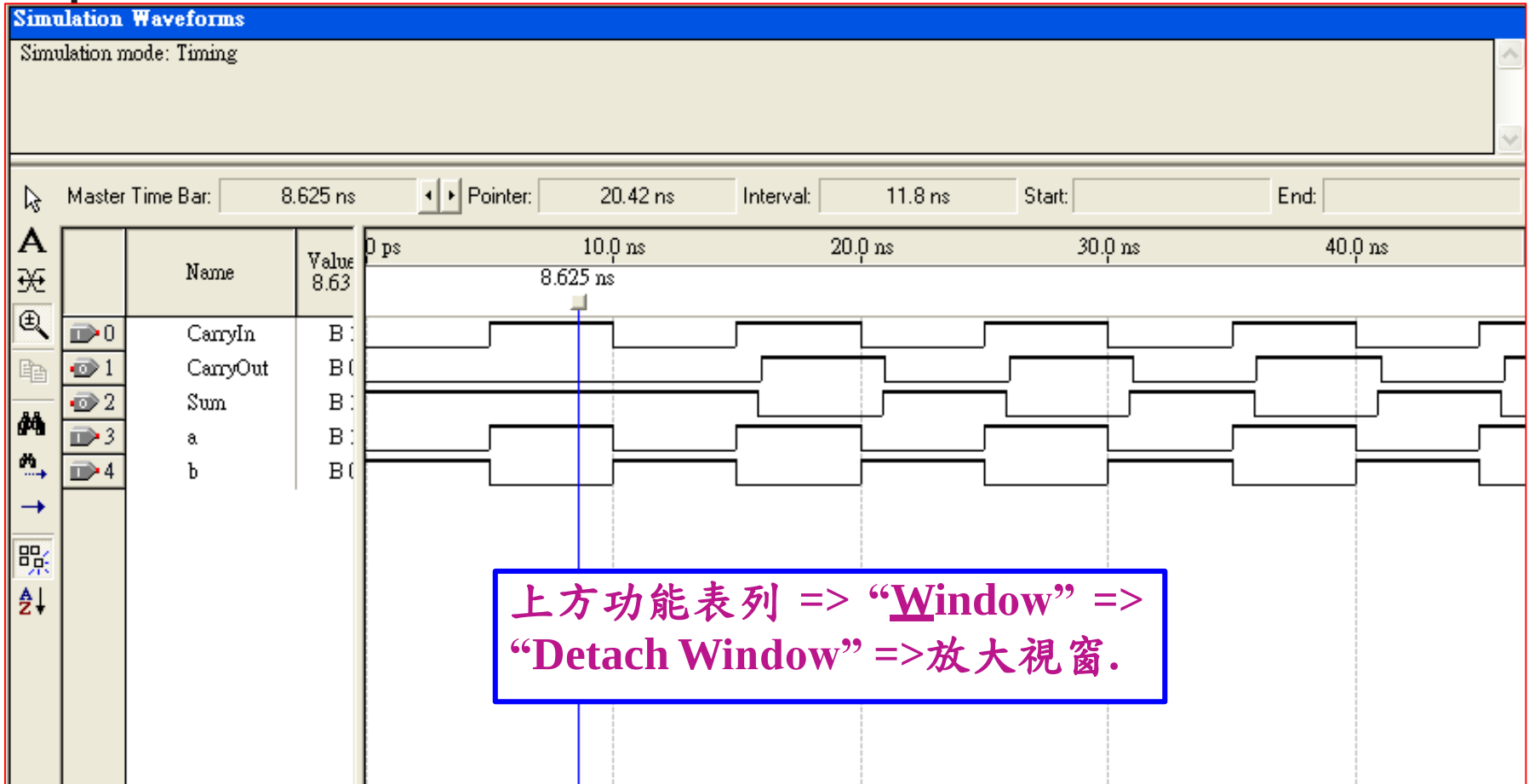
Quartus II 使用練習 (21)



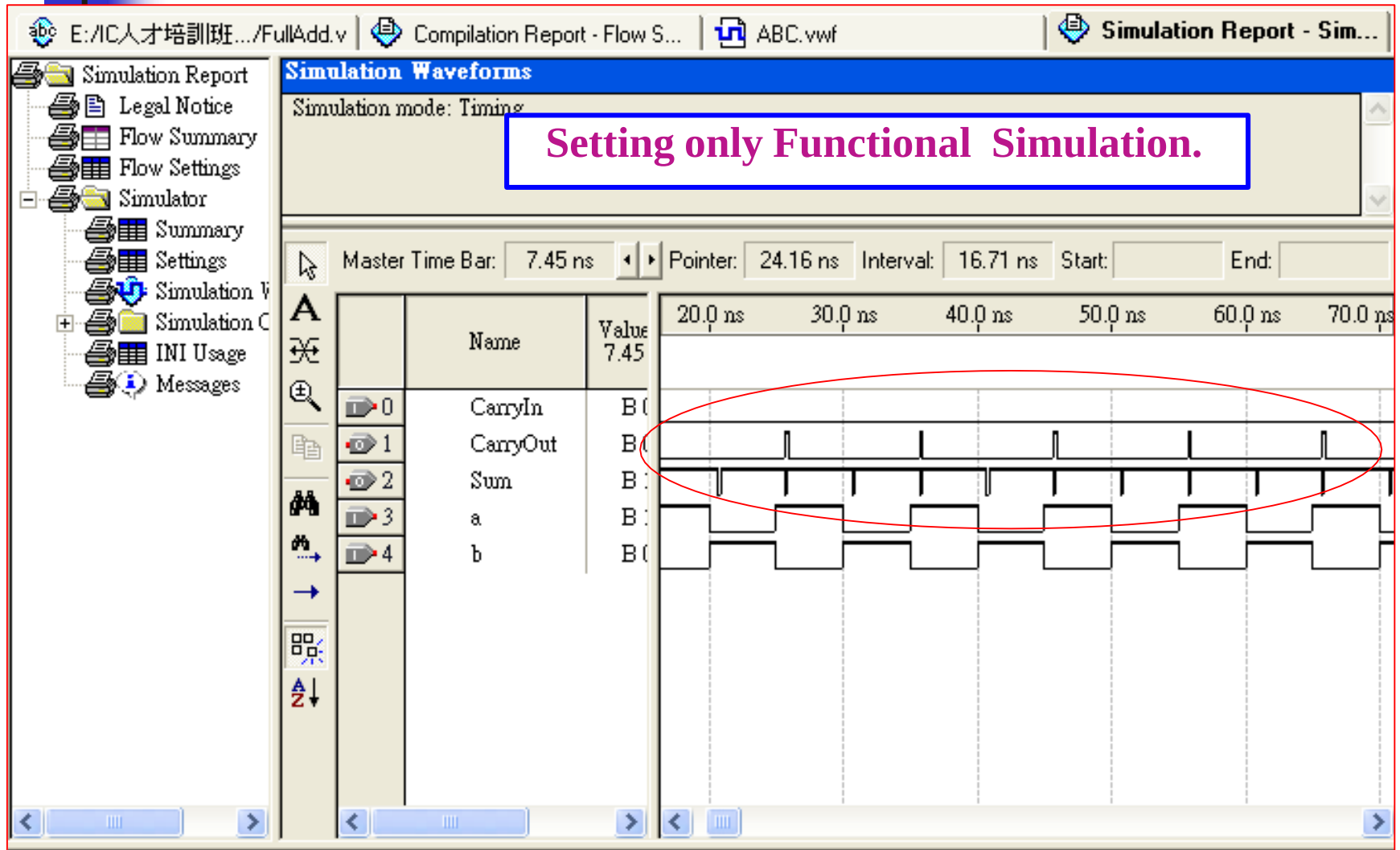
Quartus II 使用練習 (22)



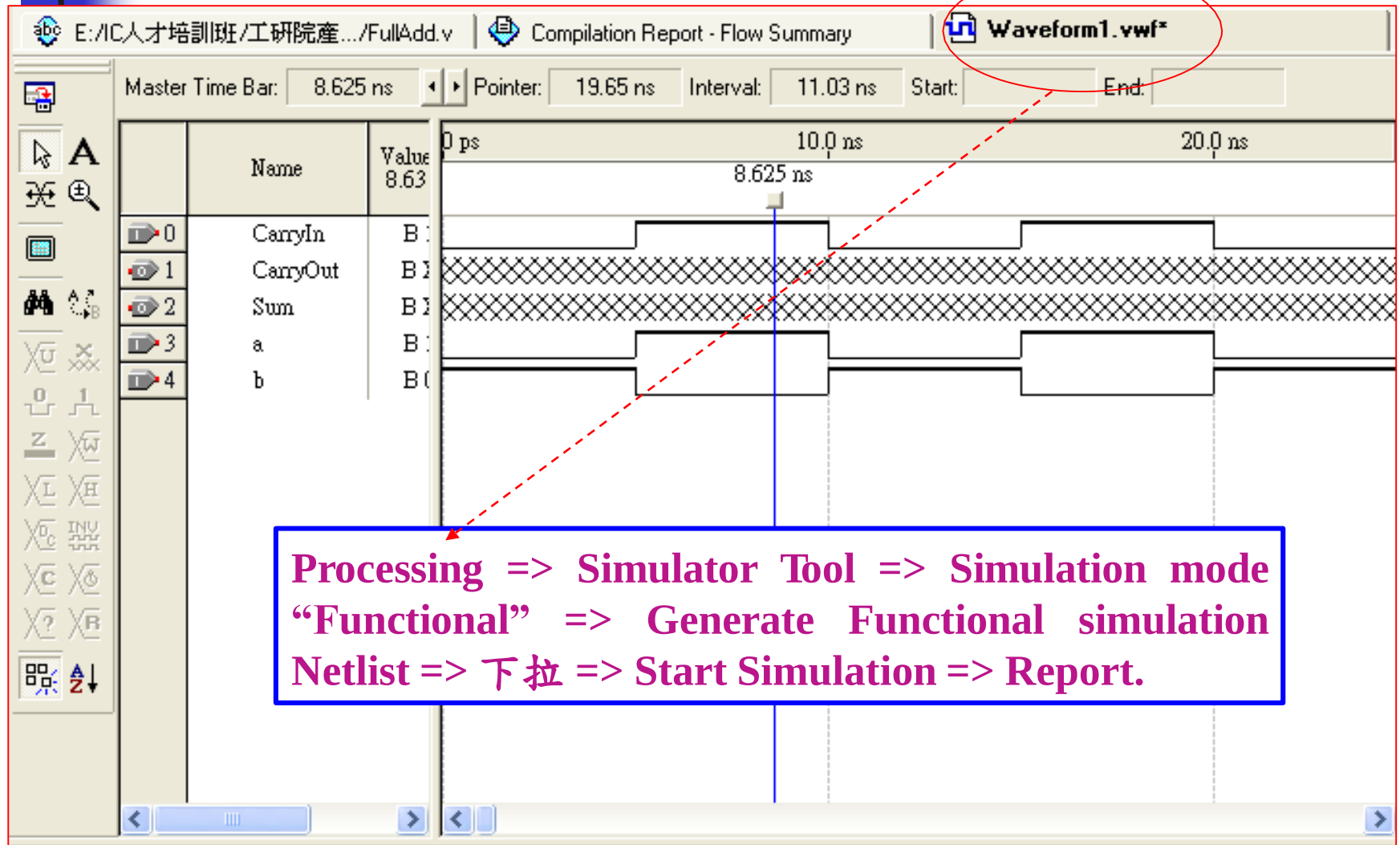
Quartus II 使用練習 (23)



Functional mode Setting (1)



Functional mode Setting (2)



Functional mode Setting (3)

The screenshot displays the Quartus II software interface. The 'Processing' menu is open, and the option 'Generate Functional Simulation Netlist' is highlighted with a red circle. Below the menu, a text box provides instructions for handling errors. The Messages window at the bottom shows the command executed and the error message received.

Processing Menu Options:

- Stop Processing (Ctrl+Shift+C)
- Start Compilation (Ctrl+L)
- Analyze Current File
- Start
- Compilation Report (Ctrl+R)
- Start Compilation & Simulation (Ctrl+Shift+K)
- Generate Functional Simulation Netlist**
- Start Simulation (Ctrl+I)
- Simulation Debug
- Simulation Report (Ctrl+Shift+R)
- Start Software Build (Ctrl+Q)
- Compile Current File

Text Box:

If there is an error, do the following procedure:
Processing => Generate Functional Simulation Netlist.

Messages Window:

```
Info: *****
Info: Running Quartus II Simulator
Info: Command: quartus_sim --read_settings_files=on --write_settings_files=off test -c test
Error: Run Generate Functional Simulation Netlist (quartus_map test --generate_functional_sim_netlist) to generate functional simulation netlist for top level entity "test" before running the Simulator (quartus_sim test)
Error: Quartus II Simulator was unsuccessful. 1 error, 0 warnings
```

Functional mode Setting (4)

The image displays the Quartus II software interface during a simulation. The main window shows the 'Waveform1.vwf*' file. The 'Simulation Waveforms' window is open, showing the simulation mode set to 'Functional'. The 'Simulation Waveforms' window displays the simulation results for 'clk_in' and 'div_2'. The 'Simulation Waveforms' window shows the simulation mode set to 'Functional'. The 'Simulation Waveforms' window displays the simulation results for 'clk_in' and 'div_2'. The 'Simulation Waveforms' window shows the simulation mode set to 'Functional'. The 'Simulation Waveforms' window displays the simulation results for 'clk_in' and 'div_2'.

Simulation Waveforms
Simulation mode: Functional

Master Time Bar: 0 ps | Pointer: 182 ps | Interval: 182 ps | Start: | End: |

Name | 0 ps | 10.0 ns

clk_in | 0 ps

div_2 | 0 ps

Simulation was successful